Agent-Mediated Search, Preference Compression, and Congestion in Matching Markets

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Abstract

As AI agents increasingly search for opportunities on behalf of people, they may fundamentally change how matching markets operate. By examining many more alternatives at little additional cost, agents can help users find better matches. Yet agents may not fully understand the preferences of the people they represent. When preference information is incomplete, agents may rely on common measures of quality and direct users with different tastes toward the same alternatives. In markets with limited capacity, these correlated choices can create congestion and offset the benefits of broader search. We study this tradeoff in a simple model and a controlled hotel-choice experiment. Our analysis shows that, despite expanding the set of opportunities users can explore, AI-mediated search can leave them worse off than searching without an agent.

1. Introduction

AI agents can examine hundreds of job listings, homes, hotels, and other opportunities without requiring users to evaluate each option themselves. The marginal cost of checking another alternative can therefore be very low. These capabilities could substantially improve matching in markets where search has historically been costly.

Finding more options, however, is not the same as knowing which option a person prefers. Users cannot always communicate their preferences fully to an agent. Some tastes are difficult to articulate; others become apparent only after seeing particular options. A traveler may have strong views about a hotel's atmosphere or layout without being able to specify those views in advance. An agent can therefore search more extensively than its user while still lacking information that matters for the user's eventual satisfaction.

This limitation has implications beyond the quality of any one recommendation. Consider a market in which agents observe the same set of hotels and receive only brief descriptions of their users. In the absence of detailed personal information, agents may place disproportionate weight on characteristics they can evaluate for everyone, such as guest ratings or hotel class. Travelers with different underlying tastes can then receive similar recommendations. We call this *preference compression*: heterogeneous preferences are represented by rankings that are more alike than the users' true rankings.

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When options are scarce, similar recommendations create an allocation problem. A hotel has a finite number of rooms; an employer has a limited number of vacancies; an appointment provider has a fixed schedule. If agents direct many users toward the same attractive option, some users are rejected while other options remain available. The issue is not that agents fail to discover alternatives. Rather, incomplete personalization makes their search decisions correlated, and capacity constraints turn those decisions into congestion.

We study this interaction in a two-option model and in a controlled language-model experiment. In the model, users differ in their valuation of match-specific characteristics. Agents can inspect both options, but receive only an attenuated signal of those preferences. We identify when agents send users with different preferred matches to the same higher-common-quality option. In that region, tighter capacity increases rejection and can reverse the welfare advantage of broader search.

For empirical evidence, we construct hotel-choice tasks with known underlying preferences. Claude ranks alternatives for synthetic travelers under progressively richer descriptions of their tastes. We compare the concentration of agent recommendations with the concentration that would arise if each traveler chose their truly preferred hotel. With limited personalization, the agents select a much narrower set of hotels; additional user-specific information disperses their choices. When capacities are imposed on those decisions, the concentrated recommendations cause rejection even though the market could accommodate every traveler in aggregate. An information-constrained Bayesian benchmark shows that the same qualitative mechanism can arise without assuming that agents make irrational use of the signals they receive.

The next section situates the paper in the literature. Sections 3–5 develop the model, experimental design, and results. Section 6 concludes. Auxiliary analyses and robustness exercises are in the appendix.

2. Related Literature

Our starting point is the distinction between expanded search and imperfect representation. Liang (2026), in *Artificial Intelligence Clones*, studies how representations of individuals can make large-scale search possible while introducing errors that persist even when many potential matches are examined. We take the difficulty of representing preferences as given and examine a different consequence: when many agents rely on similar incomplete information, their decisions may become correlated across users.

Anthropic's *Project Swap* provides an applied motivation. Participants described their preferences to language-model agents that subsequently traded books on their behalf. The agents' inferred rankings differed substantially from participants' own rankings, and imperfect preference representation accounted for a large share of the measured efficiency loss (Anthropic, 2026). Related work studies the elicitation of preferences with language models (Li et al., 2025). These studies underscore that communicating preferences to an agent is an economic constraint, not simply a matter of improving the search interface. Our emphasis is on how this constraint affects the distribution of demand and the use of capacity.

A second connection is algorithmic monoculture. Kleinberg and Raghavan (2021) show that the widespread use of a common algorithm can correlate decisions and affect social welfare. In our setting, distinct agents represent different principals, but common information about the alternatives can nevertheless make their recommendations converge. The compression of heterogeneous preferences is the source of this convergence.

Finally, the paper relates to work on the design of search and matching platforms. Information about availability, acceptance, and the characteristics of potential matches can shape where users search and whether attempts lead to transactions (Fradkin, 2017; Romanyuk and Smolin, 2019). We focus on a setting in which search across options is inexpensive, but learning user-specific fit and coordinating applications remain difficult. The resulting congestion is generated by the agents' information structure rather than by their inability to inspect the market.

3. Model

3.1 Preferences and imperfect personalization

There are N users divided equally between two types, a and b, and two alternatives, A and B. User i's utility from alternative j is

$$u_{ij} = q_j + v_{ij}.$$

The term q_j is the value of **common quality** at alternative j, which is the same for both types. The term v_{ij} is the value of that alternative's **fit** for user i. Alternative A has common quality $q_A = \Delta > 0$, while $q_B = 0$. The parameter Δ therefore measures A's advantage in common quality. Each user type gains an additional $\theta > 0$ from one particular alternative: $v_{aA} = v_{bB} = \theta$, while $v_{aB} = v_{bA} = 0$. Thus, the two types have utilities

User type	Alternative A	Alternative B
a	$\Delta + \theta$	0
b	Δ	θ

Type a always prefers A. Type b prefers B if $\theta > \Delta$: the advantage from personal fit exceeds A's common-quality advantage.

An agent examines both alternatives but recognizes only a fraction $\rho \in [0, 1]$ of a user's personal-fit value. Its estimated utility is

$$\widehat{u}_{ij} = q_j + \rho v_{ij}.$$

Here ρ measures how well the agent represents individual fit; it is a theoretical parameter, not an estimate from the experiment. The agent recommends A to type b when $\Delta > \rho\theta$. Therefore, both types are directed to A, despite preferring different alternatives, exactly when

$$\boxed{\rho\theta < \Delta < \theta.}$$

Proposition 1 (preference compression). Under this condition, agents recommend the same alternative to users who truly prefer different alternatives. The correlation comes from incomplete information about fit, not from a failure to search the available options.

For later use, define $r = \Delta/\theta$, the ratio of the common-quality advantage to the preferred-match fit advantage. The compression condition is simply $\rho < r < 1$. When r is small enough that $r < \rho$, even an imperfect agent recognizes type b's preference for B, and compression disappears.

3.2 Capacity and the welfare consequence

Each alternative can serve at most c users. Define $m = N/(2c)$ as total demand divided by total capacity. We consider $1/2 \leq m \leq 1$. At $m = 1/2$, either alternative can serve every user on its own; at $m = 1$, each can serve half of the users, and the market has exactly enough capacity in total. Larger m means tighter capacity. Balanced applications can all be served throughout this range, but concentrated applications may be rejected.

To represent the advantage of more extensive search, suppose humans know their own preferences but find their truly preferred alternative with probability $s \in [0, 1]$. Otherwise, they apply to the other alternative. Because the two user types are equally numerous, applications are balanced. Average human welfare per user is

$$W_H = \frac{\Delta}{2} + s\theta.$$

An agent that correctly distinguishes the two user types sends each to its preferred alternative. Average welfare is then

$$W_A^{high} = \frac{\Delta}{2} + \theta,$$

which exceeds human-search welfare by $(1 - s)\theta$. This is the potential benefit of broad search combined with sufficient personalization.

Under preference compression, however, all users apply to A. Because A can accommodate only c of the N applicants, random rationing gives each applicant an acceptance probability $c/N = 1/(2m)$. Rejected applicants do not apply again in this baseline model, and their utility is normalized to zero. Average welfare becomes

$$W_A^{low}(m) = \frac{1}{2m}\left(\Delta + \frac{\theta}{2}\right).$$

The expression in parentheses is the average utility from A across the two types; the factor $1/(2m)$ is the probability of admission. Thus, preference compression determines where users apply, and capacity determines how many of them are served.

To see when the welfare loss from concentrated applications outweighs the search advantage, subtract human welfare:

$$W_A^{low}(m) - W_H = \frac{\theta}{4m}\left[2(1-m)r + 1 - 4ms\right], \qquad r = \frac{\Delta}{\theta}.$$

The leading factor is positive, so compressed agent search has lower welfare precisely when the expression in square brackets is negative. For $m < 1$, solving that inequality for r gives

$$\boxed{r < \frac{4ms - 1}{2(1-m)}}.$$

At $m = 1$, the welfare difference instead simplifies to $\theta(1/4 - s)$; no division by $1 - m$ is needed. If $s > 1/4$, compression always reduces welfare at $m = 1$.

Proposition 2 (capacity amplifies compression). Holding recommendations fixed in the compression region, tighter capacity reduces welfare. When $s > 1/4$, it also expands the range of relative common-quality advantages r for which compressed agent search is worse than human search.

Appendix B illustrates the compression and welfare-loss regions. The result is conditional on preference compression: if the agent recognizes sufficiently much personal fit, it preserves different users' preferred alternatives and avoids concentrated applications.

4. Controlled Hotel-Choice Experiment

4.1 Environment and preference signals

We construct a fixed menu of 20 real San Francisco hotels, described by reference prices, hotel class, review scores, and breakfast offerings. Hotel names are hidden in the primary experiment. We generate 30 synthetic travelers with continuously heterogeneous preferences and known utility over the hotel menu.

For each traveler i and hotel j, ground-truth utility has the **same additive form as in the model**:

$$u^*_{ij} = q_j + v_{ij}.$$

The common-quality and traveler-specific components are both linear:

$$q_j = \alpha_0 + \alpha_1 \,\text{Rating}_j,$$

$$v_{ij} = \beta_{0i} - A_i \log(\text{Price}_j) + B_i \,\text{Stars}_j + D_i \,\text{Breakfast}_j + S_i \epsilon_{ij}, \qquad \epsilon_{ij} \sim N(0,1).$$

Here Rating_j is hotel j's guest rating, Price_j its nightly price, Stars_j its hotel class, and Breakfast_j its breakfast score. The coefficients α_0 and α_1 are common to all travelers, so everyone receives the same q_j for a given hotel. The coefficients β_{0i}, A_i, B_i, D_i, and S_i vary by traveler: they determine the traveler's fit with each hotel. The shock ϵ_{ij} captures additional hotel-specific taste. We draw the underlying traveler coefficients from uniform distributions and transform them into the linear coefficients above; Appendix C explains the construction and its scaling. The numerical coefficients are never disclosed to the language model.

For each traveler, the researcher converts the independently drawn preference coefficients into **low, medium, or high** categories using fixed thresholds. Each category corresponds to a natural-language description: for example, travelers with different price-sensitivity draws may be described as caring relatively little about price, caring moderately, or being strongly price-conscious. Consequently, **the wording varies across travelers**. Claude receives these descriptions, never the numerical coefficients. Each successive treatment adds information about the same traveler, while the hotel attributes remain unchanged. The precise category thresholds are reported in Appendix C.

Treatment	Information supplied about the traveler
T0	No traveler-specific information
T1	Coarse verbal description of price sensitivity
T2	T1 plus a coarse verbal description of preference for hotel class
T3	T2 plus a coarse verbal description of breakfast preference
T4	T3 plus the IDs of the three highest and three lowest idiosyncratic taste draws for that traveler, without revealing the draw values

To show what these treatments mean in practice, consider traveler U01 from the experiment. The following text was supplied as the treatments accumulated:

- **T0:** "No traveler-specific preference information is available."
- **T1 adds:** "Price matters moderately; the traveler balances cost against other hotel attributes."
- **T2 adds:** "The traveler places substantial value on a higher hotel star/service level."
- **T3 adds:** "The traveler moderately values having a good breakfast option."
- **T4 adds:** "Especially positive private affinity: H05, H13, H07" and "Especially negative private affinity: H11, H14, H08."

How T4 is constructed. For every traveler–hotel pair, the simulation draws an independent idiosyncratic taste shock $\epsilon_{ij} \sim N(0,1)$. We multiply that draw by the traveler's positive taste-scale coefficient S_i in the utility equation above. For each traveler, we select the three hotels with the largest values of $S_i\epsilon_{ij}$ and the three with the smallest values. Since S_i is positive and constant across hotels for a given traveler, these are also the top three and bottom three ϵ_{ij} draws. We tell Claude which hotels they are, using the quoted **"Especially positive private affinity"** and **"Especially negative private affinity"** statements. Claude does not see the numerical shocks. **These are not the traveler's three best and worst hotels overall:** total utility also depends on guest rating, price, hotel class, and breakfast. This strong, synthetic information signal is intended to test whether revealing otherwise unobserved personal tastes reduces correlated recommendations.

In every treatment, the model sees the same 20 anonymized hotel IDs and the same listed prices, hotel classes, guest ratings, and breakfast attributes. It must return a complete best-to-worst ranking and its first choice. Each traveler–treatment combination is run with two randomized menu orders, for 300 baseline API calls to the requested model identifier `claude-sonnet-5-5`. The prompts, preference draws, and output logs are retained for replication.

4.2 Measuring concentration

Our primary outcome is the concentration of first-choice recommendations across travelers. Let s_j be the share of agent recommendations directed to hotel j. We report

$$H = \sum_{j=1}^{20} s_j^2,$$

the Herfindahl concentration index: it is larger when more users choose the same few hotels. We compare this index with the corresponding concentration of users' true utility-maximizing first

choices. We also measure ranking accuracy and regret, defined as the loss in true utility from following the agent's top recommendation instead of the traveler's own best choice.

Treatment	Agent concentration H	True-choice H	Pairwise ranking accuracy
T0	0.668	0.133	0.636
T1	0.296	0.133	0.644
T2	0.280	0.133	0.658
T3	0.214	0.133	0.674
T4	0.172	0.133	0.747

Recommendations are substantially more concentrated than travelers' true first choices when the agent lacks individual preference information. With progressively richer signals, the distribution of recommendations becomes more dispersed and approaches the distribution based on true preferences. Pairwise ranking accuracy also improves, indicating that the additional information helps the agent distinguish travelers with different tastes. Further accuracy and regret results appear in Appendix C.

4.3 An information-constrained Bayesian benchmark

Why are recommendations so concentrated when preference information is sparse? One possibility is that the language model fails to use the information it receives. Another is that **even an agent that uses the information correctly** would make similar recommendations for different travelers because their individual tastes remain largely unknown. We construct a Bayesian benchmark to separate these explanations.

The benchmark knows the distributions used to construct the weighted common-quality and individual-fit components, including the hotel-specific taste shocks and observed hotel attributes. It does **not** observe any traveler's numerical preference coefficients or realized idiosyncratic tastes. It sees only the same signals made available to Claude under each treatment. For each traveler and treatment, it computes the expected true utility of every hotel conditional on the revealed information, then ranks the hotels by those expected utilities.

For example, T1 describes a traveler as moderately price-sensitive. The benchmark uses the distribution of price coefficients consistent with that category rather than the traveler's exact coefficient. T2 and T3 additionally restrict the distributions of the hotel-class and breakfast coefficients. T4 identifies three high- and three low-affinity hotels without revealing their latent taste values. The benchmark incorporates this ordinal information using expected order statistics under the assumed normal taste-shock distribution; Appendix C describes this approximation.

This is an **information-constrained** comparison, not a full-information optimum. The benchmark uses the available signals under the specified preference model, but its predictions can still depart from travelers' true choices because the signals omit relevant personal information.

The benchmark's first-choice concentration is 1.000 in T0 and falls to 0.151 in T4. In T0, every traveler has the same observable information, so the benchmark recommends the same hotel to everyone. As the signals distinguish travelers, its recommendations become more dispersed. The Bayesian benchmark shows that concentrated recommendations can arise simply because agents lack sufficient information about individual preferences, even when they use the available information

optimally. Claude may also make errors in interpreting or using that information, but such errors are not necessary for preference compression to occur. Appendix C provides the full benchmark comparison and additional ranking diagnostics.

5. Congestion in the Hotel Market

We impose a capacity of c rooms at each of the 20 hotels and treat each top recommendation as an application. With N travelers, the capacity-tightness ratio is $m = N/(20c)$: total demand divided by total hotel capacity. In the initial capacity exercise, each of the 30 traveler types is replicated ten times to obtain $N = 300$. We also repeat the one-shot calculation using the separate robustness sample of 300 genuinely distinct travelers.

Each traveler applies to the top-ranked hotel. A hotel with excess demand randomly accepts applicants up to capacity; rejected users are initially unmatched. For this exercise, hotel utility is normalized by subtracting the user's lowest hotel utility, and the unmatched outside option is set to zero. This is a maintained welfare normalization rather than an empirically estimated value of being unmatched.

Consider $m = 0.20$, where aggregate hotel capacity is five times the number of travelers. In the original capacity exercise, 55.0% of users are rejected under T0, compared with 1.7% under T4 and none under true first choices. With 300 distinct travelers, the corresponding rejection rates are 54.7%, 1.3%, and zero. The concentration generated by incomplete personalization therefore creates rejection even when the market has abundant aggregate capacity.

Applications based on	Rejection rate, 300 distinct travelers	Mean normalized welfare
No user information (T0)	0.547	0.763
Richest signal (T4)	0.013	2.207
True first choices	0.000	2.679

The comparison is a mechanical capacity exercise conditional on the observed rankings, not an estimate of equilibrium responses to congestion information. The same economic result also survives a fallback rule in which rejected users apply to the next hotel on their ranking until matched: unmatchedness disappears, but more concentrated rankings generate additional applications and worse final sorting. Full capacity grids, fallback results, and the centralized assignment benchmark are in Appendix D.

6. Conclusion

AI agents have the potential to transform how people search for jobs, housing, travel, and other opportunities. By making it inexpensive to examine additional options, they can help users discover alternatives that would otherwise be overlooked. Yet broader search does not necessarily lead to better matching. Agents may still have an incomplete understanding of the preferences of the people they represent, particularly when those preferences are difficult to communicate.

This paper studies two opposing effects of agent-mediated search. First, examining more alternatives allows agents to identify better matches, potentially increasing welfare. Second, when agents have limited information about individual preferences, they may rely on common measures of quality and direct users with different tastes toward the same alternatives. In markets with limited

capacity, these correlated decisions can create congestion and reduce welfare, offsetting the gains from broader search.

We characterize this tradeoff in a simple model and examine its mechanism using a controlled hotel-choice experiment. The experiment finds that agents with limited preference information concentrate their recommendations on a small set of hotels, even though travelers' true preferences are more diverse. Providing richer information about individual tastes reduces this concentration and the resulting congestion losses.

As AI agents become more widely used, the ability to search extensively may become less of a constraint than the ability to represent individual preferences accurately. Our findings suggest that the benefits of agent-mediated search will depend not only on what agents can discover, but also on how well they understand the users they represent.

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Appendix A. Robustness exercises

This appendix reports three robustness exercises using the same requested Anthropic model identifier, `claude-sonnet-5-5`, and the same T0–T4 personalization ladder as the baseline experiment. All 2,400 additional API calls completed successfully. The robustness archive stores exact prompts and hashes, raw responses, generated users and hotel sets, menu-order permutations, parsed rankings, and token usage.

A.1 Scaling to 300 distinct travelers

The baseline ranking experiment uses 30 synthetic travelers observed under two independently randomized menu orders. To tighten sampling uncertainty in the cross-user concentration measures, we construct a larger sample of 300 distinct travelers. The original 30 travelers and their latent tastes over the original 20 hotels are preserved exactly; 270 new travelers are drawn from the same coefficient distributions in the preference-coefficient distributions in Section 4, with new independent latent taste draws. We keep the 20-hotel anonymized menu and the same five information treatments. Each traveler receives one independently randomized menu order, yielding $300 \times 5 = 1{,}500$ new calls.

Treatment	Agent HHI	Excess HHI	Top-1 accuracy	Pairwise accuracy	Mean regret
T0	0.654	0.524	0.077	0.634	0.904
T1	0.272	0.142	0.117	0.657	1.021
T2	0.232	0.102	0.130	0.675	0.888
T3	0.203	0.073	0.157	0.706	0.735
T4	0.149	0.020	0.333	0.763	0.441
Ground truth	0.130	0	–	–	0

The larger sample reproduces the baseline result closely. With no personalization, choices behave as if only $1/0.654 = 1.53$ equally popular hotels were being selected; under T4 the effective number rises to $1/0.149 = 6.70$. Pairwise accuracy also rises steadily with richer information. Mean regret is not monotone at every intermediate treatment, so we continue to treat the concentration gradient as the primary outcome.

A traveler-level bootstrap resampling the 300 distinct users gives the following 95% intervals for excess HHI:

Treatment	95% bootstrap interval for excess HHI
T0	[0.460, 0.590]
T1	[0.113, 0.177]
T2	[0.071, 0.136]
T3	[0.044, 0.103]
T4	[-0.005, 0.046]

Thus excess concentration is precisely positive in the weak- and intermediate-personalization treatments and statistically indistinguishable from zero under the richest signal.

A.2 Capacity with 300 distinct travelers

We repeat the one-shot capacity exercise directly on the 300 distinct travelers rather than replicating 30 types. Capacity per hotel is unchanged from Section 5: $c \in \{300, 150, 75, 30, 15\}$, corresponding to $m \in \{0.05, 0.10, 0.20, 0.50, 1.00\}$.

Rejection rates

m	T0	T1	T2	T3	T4	True-choice benchmark
0.05	0.000	0.000	0.000	0.000	0.000	0.000
0.10	0.297	0.000	0.000	0.000	0.000	0.000
0.20	0.547	0.163	0.123	0.077	0.013	0.000
0.50	0.723	0.530	0.483	0.403	0.293	0.240
1.00	0.827	0.747	0.683	0.623	0.507	0.527

Normalized one-shot welfare

m	T0	T1	T2	T3	T4	True-choice benchmark
0.05	1.775	1.658	1.791	1.944	2.238	2.679
0.10	1.226	1.658	1.791	1.944	2.238	2.679
0.20	0.763	1.443	1.617	1.802	2.207	2.679
0.50	0.449	0.848	0.980	1.225	1.631	2.002
1.00	0.281	0.463	0.612	0.788	1.153	1.230

The headline result is nearly unchanged. At $m = 0.20$, aggregate capacity is five times demand, yet concentrated T0 applications reject 54.7% of travelers. T4 rejects 1.3%, and true first choices generate no rejection. The baseline congestion result is therefore not mechanically driven by replicating a small number of traveler types.

A.3 Real hotel names

The baseline prompts anonymize hotel identities to keep pretrained brand knowledge from entering the ranking. We rerun the exact original 30-user, 20-hotel design and the same two menu orders while revealing each real hotel name next to its H01–H20 identifier. All displayed structured attributes and all T0–T4 traveler signals remain unchanged.

Treatment	Anonymous HHI	Named HHI	Named pairwise accuracy	Named mean regret
T0	0.668	0.493	0.647	1.025
T1	0.296	0.293	0.643	1.227
T2	0.280	0.197	0.669	1.030
T3	0.214	0.231	0.684	0.951
T4	0.172	0.162	0.759	0.500

Revealing names changes some rankings and reduces concentration substantially in T0, but it does not eliminate the mechanism: named-hotel HHI still falls from 0.493 in T0 to 0.162 in T4, while

the true-choice HHI remains 0.133. The result therefore does not depend on stripping the model of all pretrained information about the alternatives.

A.4 Larger choice sets

We next expand the anonymized menu to $J = 50$ and $J = 100$ while retaining the original 30 travelers and two menu orders. The original 20 real hotels are held fixed. Additional alternatives are controlled synthetic extensions generated by bootstrapping empirical attribute bundles from the original hotels and applying small pre-specified perturbations. The $J = 50$ set is nested inside $J = 100$: shared alternatives have the same attributes, latent taste draws, and relative randomized ordering. The T4 elicitation budget remains fixed at three especially positive and three especially negative latent-affinity hotels, so preference information does not mechanically scale with menu size.

Treatment	HHI, $J = 20$	HHI, $J = 50$	HHI, $J = 100$
T0	0.668	0.936	0.967
T1	0.296	0.269	0.259
T2	0.280	0.146	0.283
T3	0.214	0.191	0.276
T4	0.172	0.176	0.237
Ground truth	0.133	0.136	0.102

The striking result is T0: when every user supplies the same information, a larger opportunity set can make the agents converge almost completely on a common standout option. The largest single hotel receives 96.7% of T0 top choices at $J = 50$ and 98.3% at $J = 100$. This increase is not mirrored in true preferences, whose HHI is 0.136 and 0.102 in the two expanded menus.

We do not find evidence that this concentration is driven by simple list-position bias. Averaging across the five treatments, the selected top option lies at 49.3% of the way through the displayed list for $J = 50$ and 52.4% for $J = 100$. The share of top choices drawn from the first 10% of the list is 10.3% in both cases. The mean Kendall correlation between input position and final output rank is 0.013 for $J = 50$ and 0.004 for $J = 100$.

The larger-menu exercise should nevertheless be interpreted cautiously. The synthetic expansion increases the chance that the menu contains an option that looks especially strong on broadly valued displayed attributes, and the amount of T4 hotel-specific information is held fixed while the opportunity set grows. The exercise therefore supports the mechanism that broad search can amplify coordination on common signals when personalization does not scale with the menu; it is not a clean causal test of psychological choice overload.

Appendix B. Phase diagrams

The three panels plot the theoretical model in terms of $r = \Delta/\theta$ (horizontal axis) and ρ (vertical axis), holding the human discovery probability at $s = 1/2$. These are the only two preference-information dimensions required to show the compression mechanism. The region below the line $\rho = r$, for $0 < r < 1$, contains **compressed agent rankings**. The shaded subset also satisfies the welfare-loss condition in Section 3.2.

At $m = 0.50$, each alternative can accommodate all users. Within the interior of the compression region, the search advantage is sufficient to prevent a strict welfare loss for $s = 1/2$.

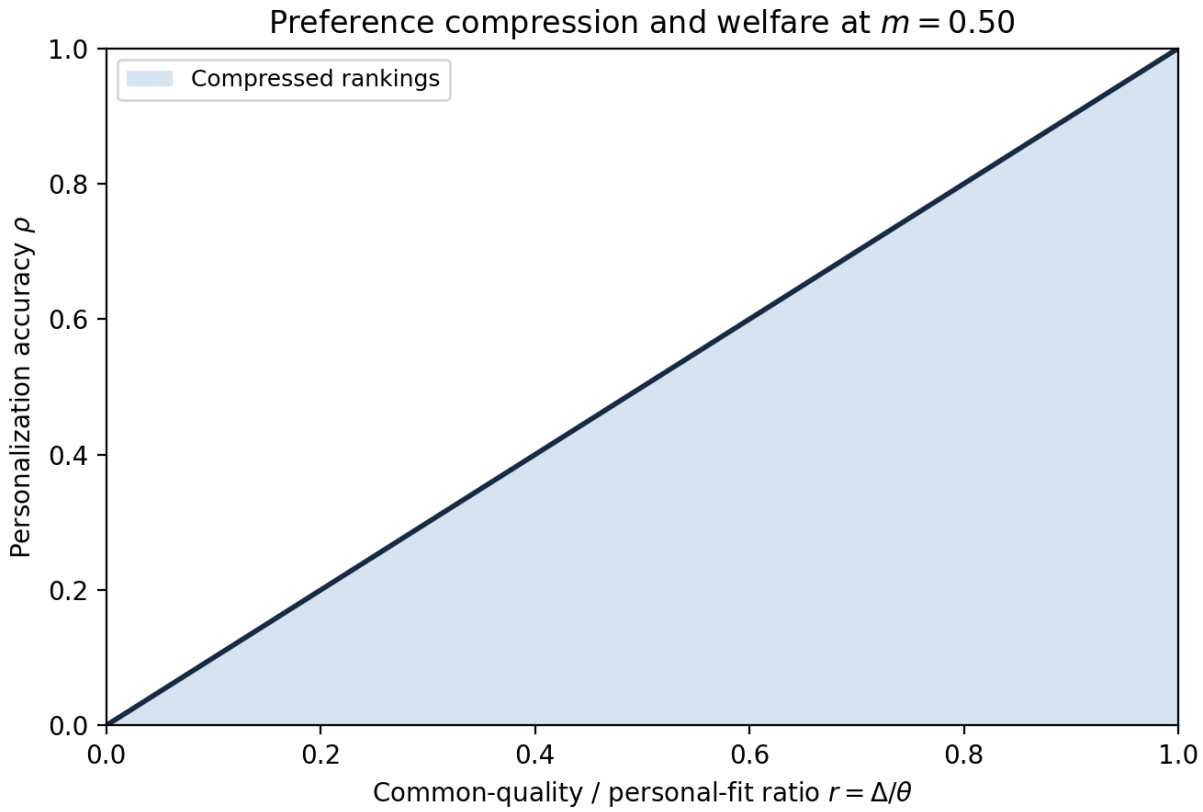


Figure 1: Compression and welfare loss with loose capacity

At $m = 0.65$, compressed rankings reduce welfare only if $r < 3/7 \approx 0.429$: personal fit must be sufficiently important relative to common quality. The agents must also satisfy $\rho < r$ to generate compression.

At $m = 1$, total capacity exactly equals total demand, but concentrated applications still lead to rejection. Because $s = 1/2 > 1/4$, every outcome in the compression region has lower welfare than human search.

The panels are theoretical illustrations, not estimates from the hotel experiment.

Appendix C. Additional Experimental Evidence

C.0 Preference coefficients and baseline results

For each traveler i, the researcher independently draws four underlying preference coefficients:

$$a_i \sim U(0.20, 0.90), \quad b_i \sim U(0, 1.50), \quad d_i \sim U(0, 1.00), \quad \sigma_i \sim U(0.25, 1.50).$$

The draws govern price sensitivity, hotel-class preference, breakfast preference, and the scale of idiosyncratic fit, respectively. For each traveler–hotel pair we also draw $\epsilon_{ij} \sim N(0, 1)$ independently.

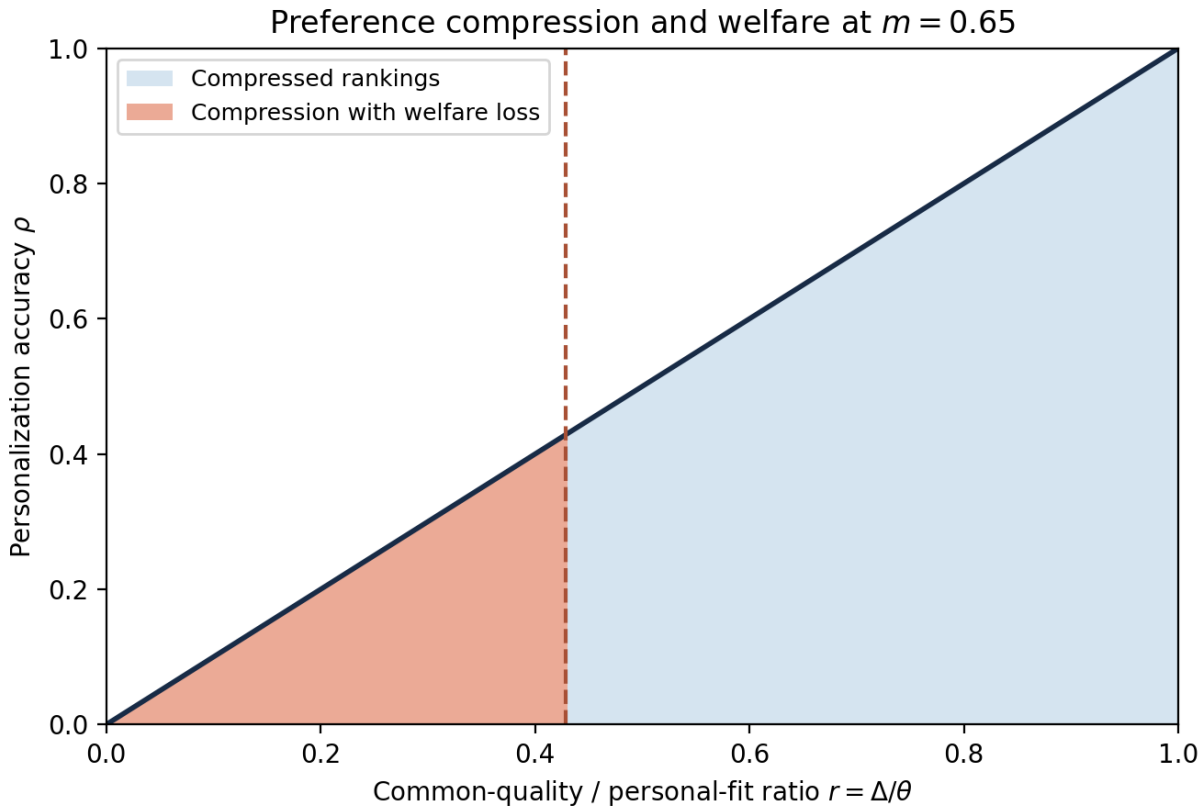


Figure 2: Compression and welfare loss with intermediate capacity

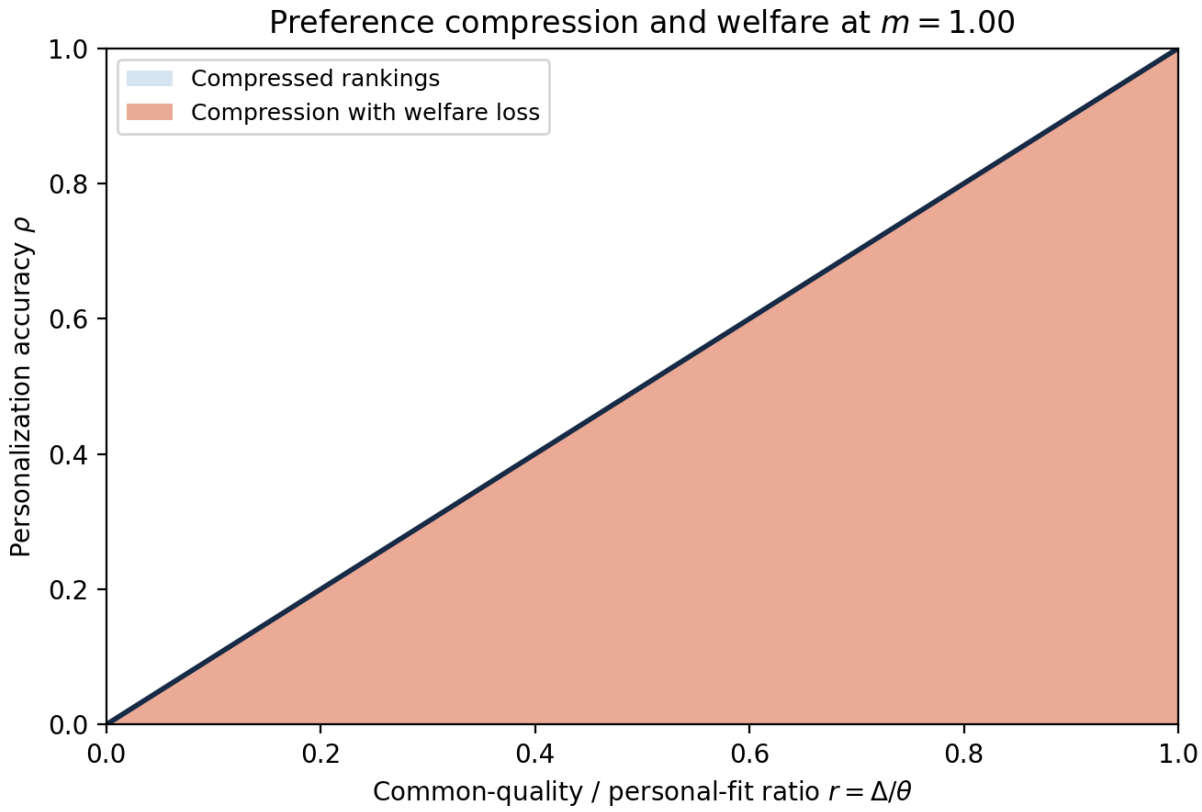


Figure 3: Compression and welfare loss with tight capacity

In the original simulation, raw fit is constructed as

$$f_{ij}^{raw} = -a_i z(\log \text{Price}_j) + b_i z(\text{Stars}_j) + d_i z(\text{Breakfast}_j) + \sigma_i \epsilon_{ij}.$$

Here $z(x)$ subtracts the average value of an attribute across the original 20 hotels and divides by its standard deviation. The simulation combines 0.30 times standardized guest rating with 0.70 times standardized raw fit (the latter standardized over the original 30 travelers and 20 hotels). Expanding these standardized expressions gives exactly the linear q_j and v_{ij} equations in Section 4.1: the rating coefficients α_0, α_1 are fixed across users, while $\beta_{0i}, A_i, B_i, D_i, S_i$ absorb the corresponding centering and scaling constants and vary with each traveler's underlying draws. The experiment's saved utilities, rankings, and welfare calculations are unchanged. The original scaling is held fixed in the robustness exercises where appropriate.

The verbal preference categories supplied to the agent are determined by the following fixed cutoffs applied separately to each traveler's original coefficient draw. Intervals are inclusive at their lower boundary and exclusive at their upper boundary, except for the final category.

Underlying coefficient	Low	Medium	High
Price sensitivity a_i	$a_i < 0.43$	$0.43 \leq a_i < 0.67$	$a_i \geq 0.67$
Hotel-class preference b_i	$b_i < 0.50$	$0.50 \leq b_i < 1.00$	$b_i \geq 1.00$
Breakfast preference d_i	$d_i < 1/3$	$1/3 \leq d_i < 2/3$	$d_i \geq 2/3$

The numerical coefficients are not included in the prompts. The corresponding low, medium, and high descriptions are used in T1 for price, added in T2 for hotel class, and added in T3 for breakfast. For T4, the six hotel IDs come from the three largest and three smallest $\sigma_i \epsilon_{ij}$ values for each traveler; the model never sees the numerical shocks. This signal is not obtained by binning a coefficient.

The complete baseline treatment summary is:

Treatment	Agent HHI	Excess HHI	Top-1 accuracy	Pairwise accuracy	Mean regret
T0	0.668	0.535	0.133	0.636	1.002
T1	0.296	0.162	0.133	0.644	1.192
T2	0.280	0.147	0.100	0.658	1.112
T3	0.214	0.081	0.133	0.674	1.023
T4	0.172	0.039	0.267	0.747	0.578
Ground truth	0.133	0	–	–	0

C.1 Ranking-loadings diagnostic

An agent's ranking may reflect both the common information it receives about hotels and the traveler's personal preferences. To study their relative roles, we estimate descriptive pairwise ranking

regressions. We retain the **unweighted standardized predictors from the original empirical analysis** so that the reported regression estimates are unchanged; these predictors are not the already-weighted utility contributions q_j and v_{ij} defined in Section 4.

Write $g_j = z(\text{guest rating}_j)$ for standardized guest rating, and split the raw personal-fit score from Section 4.1 into

$$f_{ij}^{struct} = -a_i z(\log \text{price}_j) + b_i z(\text{stars}_j) + d_i z(\text{breakfast}_j), \qquad f_{ij}^{latent} = \sigma_i \epsilon_{ij}.$$

The structured component captures price, hotel class, and breakfast preferences; the latent component captures traveler–hotel tastes that cannot be inferred from those displayed attributes. For every pair of hotels j, k and treatment $h \in \{T0, T1, T2, T3, T4\}$, we estimate

$$\Pr(j \succ_i^A k) = \Lambda\left(\alpha_h + \beta_h^g (g_j - g_k) + \beta_h^S (f_{ij}^{struct} - f_{ik}^{struct}) + \beta_h^L (f_{ij}^{latent} - f_{ik}^{latent})\right),$$

where $j \succ_i^A k$ means the agent ranks hotel j above hotel k for traveler i, and Λ is the logistic function. We report each fit coefficient relative to the guest-rating coefficient, β_h^S/β_h^g and β_h^L/β_h^g. **These are descriptive regression ratios, not the theoretical ratio $r = \Delta/\theta$ and not estimates of ρ.** Because the regressors retain the original unweighted scales, they also should not be interpreted as marginal coefficients on the weighted q_j and v_{ij} utility terms.

Treatment	Structured-fit / rating loading	Latent-fit / rating loading
T0	0.252	0.034
T1	0.611	-0.034
T2	0.811	-0.031
T3	1.437	-0.087
T4	1.511	1.536

The fit loading increases as corresponding information becomes available. Latent-fit loading is close to zero through T3 and increases when T4 reveals identities of especially high- and low-affinity hotels. The correlation between the unweighted standardized common-quality measure and total standardized fit is -0.018 in levels and -0.001 in pairwise differences, ruling out strong collinearity between those original regressors as an explanation of the pattern.

C.2 Bayesian benchmark details

The benchmark ranks hotels by posterior expected true utility $u_{ij}^* = q_j + v_{ij}$, conditioning on the hotel attributes and the traveler information available in each treatment. It knows the parameter distributions in Appendix C.0, but not any individual traveler's realized coefficients. For a coarse label such as "moderately price-sensitive," the posterior mean of the price coefficient is calculated over the coefficient range consistent with that label. Unrevealed coefficients are averaged over their prior distributions. These conditional expectations determine the benchmark's predicted fit contribution v_{ij}, including the original 0.70 weight.

In T4, the benchmark also receives the IDs of the three hotels with highest and three with lowest latent affinity, but not the magnitudes of the underlying shocks. It approximates their latent contribution using the expected ordered values of independent standard-normal draws. This is an

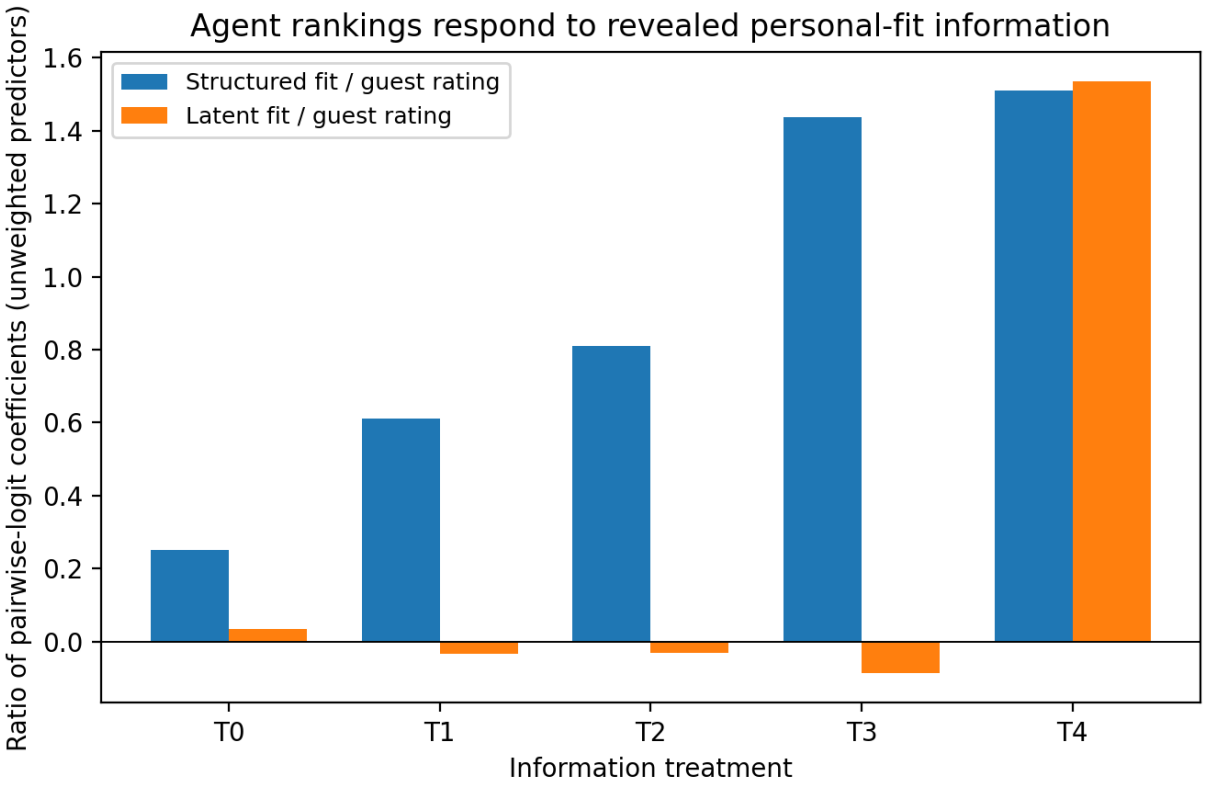


Figure 4: Agent ranking loadings on observed quality and individual fit

order-statistic approximation to the conditional information in the revealed ranking: the identities convey relative positions, not exact utility values, and the benchmark retains uncertainty about the remaining hotels.

Treatment	Bayesian top-choice concentration	Bayesian pairwise accuracy	Bayesian mean regret
T0	1.000	0.754	0.700
T1	0.336	0.762	0.695
T2	0.269	0.776	0.425
T3	0.189	0.775	0.494
T4	0.151	0.873	0.094

These numbers describe an information benchmark, not an experimental treatment assigned to a different language model. In particular, the benchmark does not observe the full true utility matrix when it forms recommendations.

C.3 Menu-order stability

Because language models can be sensitive to list position, every user-treatment pair is evaluated under two independent hotel orders. We summarize stability using Kendall's rank correlation between the two complete rankings for the same user and treatment. Kendall's $\tau = 1$ means the two rankings are identical, 0 means no systematic ordinal agreement, and -1 means exact reversal.

Average Kendall correlations are 0.845, 0.875, 0.899, 0.883, and 0.862 for T0 through T4, respectively. Thus the full rankings are highly stable to option ordering even though exact top-choice agreement is naturally lower. The decline in HHI is also present in both order replications.

A traveler-level bootstrap that preserves the two order replications gives a 95% interval for excess HHI of $[0.385, 0.649]$ in T0, $[0.059, 0.269]$ in T1, and $[0.015, 0.298]$ in T2. The intervals include zero by T3 and T4. With 30 synthetic travelers, the estimates show strong excess concentration under weak personalization and are consistent with most of that excess concentration disappearing under richer preference information.

C.4 Reproducibility

The run used the requested Anthropic API model identifier `claude-sonnet-5-5`. Temperature was not explicitly specified in the API call, so the provider default applied. The synthetic-user seed was 20260930. The replication archive stores the frozen hotel data, exact user coefficients, every latent taste draw, exact prompts, prompt SHA-256 hashes, hotel-order permutations, raw API responses, parsed rankings, token usage, and all first-pass metrics. The complete prompt text for every call is retained in `prompts.jsonl`; no prompt reconstruction is required.

Appendix D. Allocation Details

For the original capacity exercise, crowding values $m \in \{0.05, 0.10, 0.20, 0.50, 1.00\}$ imply per-hotel capacities $c \in \{300, 150, 75, 30, 15\}$ for 300 travelers.

D.0 One-shot rejection by capacity

m	T0	T1	T2	T3	T4	True-choice benchmark
0.05	0.000	0.000	0.000	0.000	0.000	0.000
0.10	0.300	0.000	0.000	0.000	0.000	0.000
0.20	0.550	0.200	0.200	0.100	0.017	0.000
0.50	0.767	0.600	0.517	0.417	0.367	0.267
1.00	0.867	0.783	0.717	0.658	0.583	0.550

At $m = 0.20$, one-shot normalized welfare is 0.784 under T0, 2.273 under T4, and 2.878 under true first choices. At $m = 1$, the corresponding values are 0.237, 0.961, and 1.242. A centralized assignment that uses the full true-utility matrix attains 2.354 at $m = 1$.

D.1 Fallback allocation

The one-shot exercise matches the analytical benchmark but could overstate the cost of congestion if rejected users can simply try another hotel. We therefore use the complete rankings returned by the model.

Fallback proceeds in rounds. Unmatched travelers apply to their highest-ranked hotel not yet tried. Each hotel fills its remaining capacity by random rationing if oversubscribed. Rejected travelers move to the next hotel in their ranking. The process continues until everyone is matched. Because total capacity is at least total demand for all $m \leq 1$, this procedure eliminates unmatchedness by construction.

The relevant loss is now **mis-sorting**, not the outside option. We measure the true-utility regret of the final assigned hotel relative to each traveler's true favorite.

m	T0	T1	T2	T3	T4	True-ranking benchmark
0.05	1.002	1.192	1.112	1.023	0.578	0.000
0.10	1.170	1.192	1.112	1.023	0.578	0.000
0.20	1.265	1.169	1.087	1.024	0.569	0.000
0.50	1.139	1.190	1.145	1.070	0.651	0.240
1.00	1.424	1.458	1.372	1.327	1.003	0.681

The fallback robustness confirms that the capacity result is not driven solely by assuming that rejected users disappear. In the tight market $m = 1$, all travelers eventually receive a hotel, yet T0 produces mean regret of 1.424 compared with 1.003 under T4 and 0.681 when travelers use their true rankings. The average number of applications is also much larger: 8.85 under T0, 4.82 under T4, and 4.56 under the true-ranking benchmark.

Fallback regret need not be monotone at every intermediate capacity: rationing can occasionally push a user away from the agent's mistaken first choice and toward an alternative closer to the user's true preference. We therefore interpret this exercise as a robustness check on the mechanism rather than as a monotone comparative static for every treatment. The important result is that weak personalization continues to generate substantially more search effort and worse final sorting in tight markets even when no traveler is left unmatched.